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Introduction

A South African medical aid scheme approached myself to do a Linear regression model for them using a medical insurance report from the US. Medical costs can be influenced by a number of variables, and the variables in this report are age, sex, smoker and region. The request for the linear regression model is to create a predictive model to see how the report can assist to predict the cost of medical insurance when the variables input are different from the report used to create the model. The following steps were taken to create the model and some of the steps done are to clean the report, find and remove nulls or empty values in columns/rows, train and evaluate the model, etc.

The first step I did was to import the python libraries. The libraries that were imported were pandas, seaborn, matplotlib and numpy.

Importing of the libraries

```
# Importiing libraries
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np
```

Figure 1: Python libraries

Exploratory Data Analysis (EDA)

From importing the python libraries, I loaded the report which was in cvs format. I looked at the data types to see which columns are integers, objects and floats, as shown in the picture below:

```
In [7]: # diplaying of the data types in the dataset
        df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       1338 non-null   int64
 1   sex       1338 non-null   object
 2   bmi       1338 non-null   float64
 3   children  1338 non-null   int64
 4   smoker    1338 non-null   object
 5   region    1338 non-null   object
 6   charges   1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
```

Figure 2: Data types

Based on the results, there are 6 independent variables which are age, sex, bmi, children, smoker, and region. GeeksforGeeks (2025) writes that independent variables can be used to influence an outcome of a dependent variable. They further explain that a dependent variable is an outcome that is observed on an experiment. In this report the dependent variable is charges.

I then checked if there were any missing values or null data in the columns and rows. Based on the results below of 0 in all the columns, there are no null or missing values in the dataset.

```
In [9]: # checking for nulls or missing information
        df.isnull().sum()

Out[9]: age         0
        sex         0
        bmi         0
        children    0
        smoker      0
        region      0
        charges     0
        dtype: int64
```

Figure 3: No nulls results

The df.nunique() syntax looks for unique values in the dataset. The results shown below shows that there are 47 unique ages, 2 sex which is either male or female and 1337 unique values out of 1338 charge values. This means that 2 clients are charged the same amount. I did this step to see if there are any unique values and how many were not unique. This is to see if model can produce better predictions and has variety of information to work with.

```
In [10]: # checking for unique values
         df.nunique()

Out[10]: age           47
         sex            2
         bmi          548
         children       6
         smoker         2
         region         4
         charges     1337
         dtype: int64
```

Figure 4: Unique values check

Further analysis of the data is shown in the picture below where it shows the actual amounts and of the int and float values and the actual information of the objects. In this step, I wanted to see the data types if all are numeric and/or there are character values which I will have to change to the format that machine learning can understand.

```
In [3]: # Loading the dataset
        df = pd.read_csv(r"C:\Users\User\Desktop\Medical_Insurance.csv")
        df

Out[3]:
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520
...	...	...	...	...	...	...	...
1333	50	male	30.970	3	no	northwest	10600.54830
1334	18	female	31.920	0	no	northeast	2205.98080
1335	18	female	36.850	0	no	southeast	1629.83350
1336	21	female	25.800	0	no	southwest	2007.94500
1337	61	female	29.070	0	yes	northwest	29141.36030

1338 rows × 7 columns

Figure 5: Medical Insurance Dataset

The data was explored and analysed, and here is the analysis of the dataset:

The dataset contains medical insurance information of clients. There are 1338 values in each column which means that this is the total number of clients. However, the total number of entries in the dataset is 9373 including the headings. The data is structured meaning its organised in columns and rows, and schema-dependent (it's in comma-separated values file type). There are no nulls or missing values in each column and some columns contains integers, floats and varchar. There are seven columns and each column represent the following:

- Age – it's the age of the clients who have taken the medical insurance. Their ages ranges from 19 to 64 years old.
- Body Mass Index (BMI) – which ranges from 15.96 to 53.13.
- Children – this shows the number of children each client has. Some have no children while others have a maximum of 5.
- Smoker – this column indicates if the client smokes or not.
- Region – there are only 4 regions in the dataset which indicate which region each client falls under. The four regions are northwest, northeast, southwest and southeast.
- Charges – these are amount that each client is charged. It is different because it takes into consideration of the information the client has provided such as age, smoker, etc.

In the dataset, there are more male clients than females. The number of males is 676 and the number of females is 662. The number of clients who are 18 years old are the highest and they are 69 clients and the lowest number of clients are aged 64 and their total count is 22 clients.

Further analysis done on the report was to format the dataset to have 2 decimals. The original dataset had more than 2 decimals in columns bmi and charges.

The step that followed was to do statistical analysis on the dataset. The screenshot below shows the results of the statistical calculation of the dataset.

```
In [8]: # statistic calculations
        df.describe()
```

Out[8]:

	age	bmi	children	charges
count	1338.00	1338.00	1338.00	1338.00
mean	39.21	30.66	1.09	13270.42
std	14.05	6.10	1.21	12110.01
min	18.00	15.96	0.00	1121.87
25%	27.00	26.30	0.00	4740.29
50%	39.00	30.40	1.00	9382.03
75%	51.00	34.69	2.00	16639.91
max	64.00	53.13	5.00	63770.43

Figure 6: Statistic results

Summary of the statistical results is that the maximum age of the clients in the dataset is age 64 and the minimum age is 18.

Visualization of the data

The histogram graph below shows the relationship between the charges and the number of clients. The graph shows a peak on the left which means that the clients that are charged less are the greatest number of clients. Also, they represent that they may have minimal medical problems or they often go for medical check-ups. The end of the graph shows a flat stretch which could represent clients who are smokers or have chronic illnesses that would drive higher medical charges.

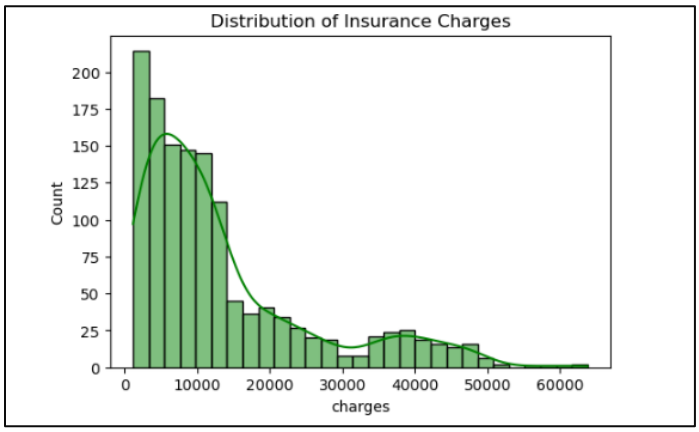

Figure 7: Distribution graph

Boxplot graph below compares the impact of smoking and sex on medical charges. The left side of the graph shows clients that smoke and are the ones that pay higher charges compared to non-smokers shown on the left side of the graph.

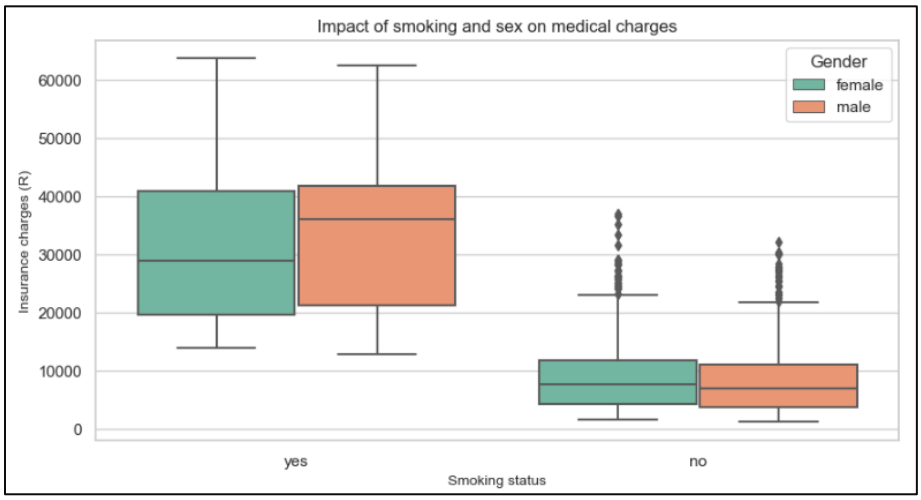

Figure 8: Boxplot graph: Smoker versus charges based on sex

In the graph below, I compared the charges against the regions for both male and female. Based on the below, both genders are charged the highest in all the regions. Southeast shows it has the highest charge on males while southwest has the least charge on females.

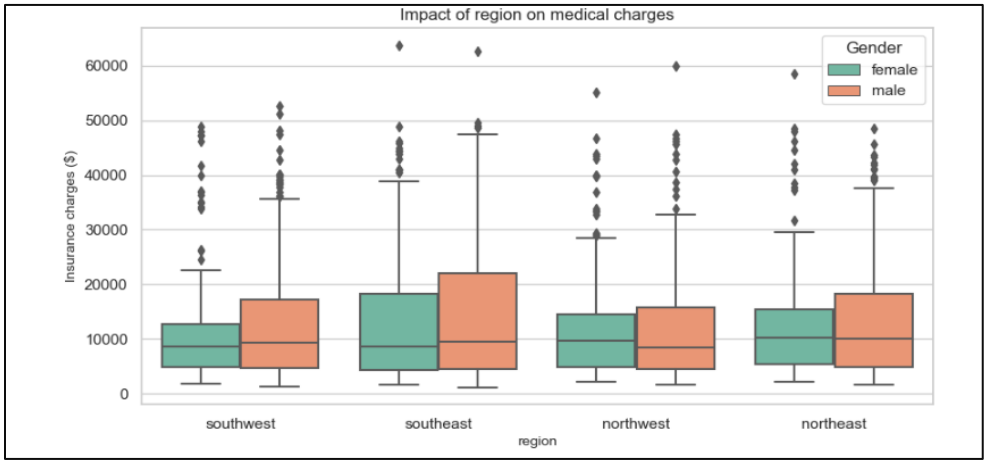

Figure 9: Boxplot graph: Region versus charges based on sex

The graph below shows how age affect medical insurance charges. This indicate that as age increases the charges also increases. Based on the results shown below, the dataset is fit to create a Linear regression model. Looking at the graph, charges that are 50000 and above are few and not scattered as the charges that are lower therefore, they were the outlier and were removed. According to GeekforGeek (2026) outliers can affect the statistical results of the "mean, standard deviation and correlation values". They further explain outliers can extend graphs which can make it difficult to provide better understating and conclusions and also machine learning models like linear regression is sensitive to values that do not form a better linear regression outcome. The outlier that I targeted and removed were on sex and region based on the charge mentioned above.

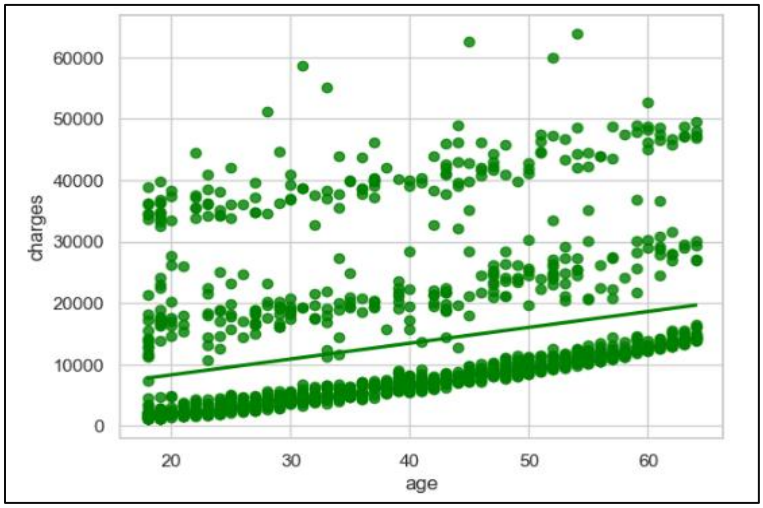

Figure 10: Scatter plot graph: Age versus Charge

Data Preprocessing

To do linear regression model some columns had to be formatted into machine language. Machine language do not understand string values therefore; sex, smoker and region columns were changed to binary (which is 1 or 0). For example, smoker column has a “yes” or “no” values therefore clients who smoke are represented by 1 and those who do not smoke are represented by 0. Male are represented by 1 and females are represented by 0. For regions since there are 4 regions, new columns were created for each region. For example, if a client is from Southwest region, the number 1 will represent a “yes” for that client.

The picture below, shows the script used to convert sex, smoker and region into the language that machines understand. Results shows the split of regions and the 1 or 0 as described above.

```
In [42]:
# Convert categorical variables into dummy variables
# drop_first=True helps avoid 'multicollinearity' (redundant data)
df_encoded = pd.get_dummies(df, columns=['sex', 'smoker', 'region'], drop_first=True)

# View the new numerical columns
print(df_encoded.head())

   age   bmi  children  charges  region_northwest  region_southeast  \
0   19 27.90         0 16884.92                 0                 0
1   18 33.77         1  1725.55                 0                 1
2   28 33.00         3  4449.46                 0                 1
3   33 22.70         0 21984.47                 1                 0
4   32 28.88         0  3866.86                 1                 0

   region_southwest
0                 1
1                 0
2                 0
3                 0
4                 0
```

Figure 11: Conversion of variables

Features and Target Variables

In this section I divided the dataset into features and target variables. Features are the information to be used to predict and the target variables are the predictions that will be predicted. The target variable is charges since I want to predict if charges will be affected by the features. Therefore; the features will be the X which will be age, bmi, smoker and region.

```
In [44]: # Assumption: 'df_encoded' the DataFrame after converting categories to numbers

# Defining the Features (X)
# Dropping 'charges' so that the model doesn't see answers in advance
X = df_encoded.drop(columns=['charges'])

# Defining the Target Variable (y)
y = df_encoded['charges']

# Verifying the shapes
print(f"Features shape: {X.shape}") # showing all colunms except the 'charges'
print(f"Target shape: {y.shape}")     # showing only one column

Features shape: (1338, 6)
Target shape: (1338,)
```

Figure 2: Defining features and target variables

Splitting the dataset into training and testing Sets

Then I proceeded to spitting the dataset 80% training and 20% for testing. Gillis, 2024 writes that in the importance of splitting the dataset in Data Science is an important characteristic when creating data models like the linear regression that is created in this assignment. He further explains that the split can be in 2 parts, one part for testing and the other part for training the model.

```
In [108]: # Spliting the data
          X_train, X_test, y_train, y_test = train_test_split(
              X_clean, y_clean, test_size=0.5, random_state=25
          )
```

Figure 12: Data splitting

Trained Linear Regression Model

The below script is for training the linear regression model as the results are shown.

```
In [109]: # Trainining the model
          model = LinearRegression()
          model.fit(X_train, y_train)

Out[109]: LinearRegression()
```

Figure 13: Trained Linear regression model

Making prediction

The below script is for making predictions and the results shows a prediction of 9 charges.

```
In [110]: # making a prediction
          y_pred = model.predict(X_test)
          print(y_pred[:9])

          [15135.49682642 12183.64125159 18846.18370357 11388.72405574
           13373.58661701  7372.42142528 15975.51186459 16202.62849985
           17963.66616395]
```

Figure 14: Predictions and results

The below scatter graphs compare the predicted versus actual charges. Geeksforgeeks (2026) state that the line seen skewed up-right is a best-fit line that is used to make predictions. They further explain that the feature and target of the linear regression model are assumed to have a relationship between them. Its goal is to minimise error (Geeksforgeek, 2026).

Therefore, based on the results I got from the model demonstrated in the picture below those dots that are directly on the line means that they are the perfect predictions for charges.

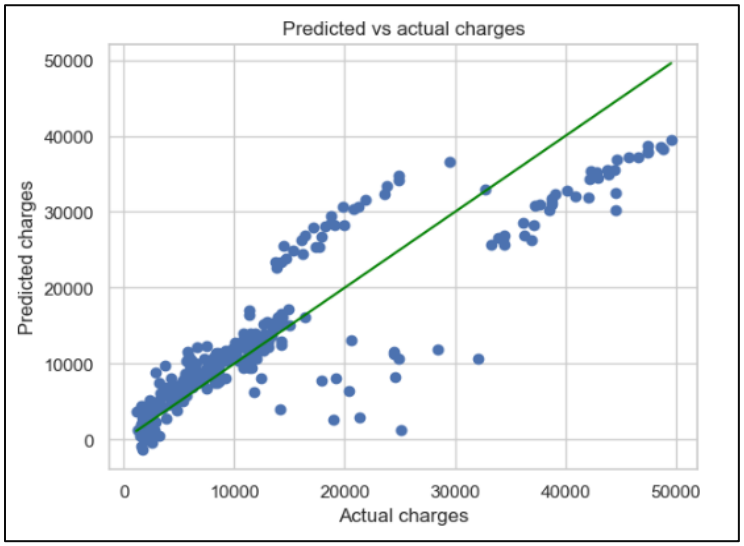

Figure 15: Scatter graph: Predicted vs actual charges

Model performance evaluation

Model performance evaluated by importing scikit-learn metrics and showing the results. Fernando (2025) emphasises that in a Linear regression model the R-squared should be read and understood as a percentage value of a dependent variable that is influenced by independent variables.

The results on the R-squared in this model is 78%(rounded-up) which is a good measure for the relationship between the model and the dependent variable.

```
# importing scikit-learn metrics
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# evaluating the model with scikit-learn metrics
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("Mean absolute error:", mae)
print("mean squared error:", mse)
print("root mean squared error:", rmse)
print("R-squared score:", r2)
```

```
Mean absolute error: 4226.40971225337
mean squared error: 35673232.89658567
root mean squared error: 5972.707333913631
R-squared score: 0.7753095020969256
```

Figure 16: Scikit-Learn metrics

Reflection

The variables that influence the medical insurance charges strongly are smokers and age. Smoking is the one that affects the insurance costs the most because it means that they are at risks of attracting more illnesses than those clients who do not smoke. Based on the graph below, it clearly shows that those who smokes their costs are higher than those who do not.

Findings and recommendations

Based on the analysis of the US dataset provided, there data was not large enough to produce a more predictability model. It also had outliers that if I didn't remove them they would have skewed the model therefore affected the findings and predictions. The dataset showed that only region, age, smoker and region are what they used to influence each client's charges. Another flaw seeing in the data is that one person can be charged above R50 000 while another one can be charger less than R10 000 and they didn't add months to indicate if its monthly charges on the whole year's charges.

My recommendation for you as the South African medical scheme is to get a larger dataset that also includes monthly charges, clients' chronic illnesses, health history including family health history, occupation of clients and alcohol drinking habits. By adding the mentioned information and larger dataset you can be able to get a model trained and it can produce better predictions for medical charges.

Conclusion

Taking into consideration the dataset used in creating this model, Linear Regression model is a good model to predict medical insurance charges. The features that were in the model such as smokers and age showed that they are some of the good features for the model. As I have mentioned in the findings and recommendations, the model can produce better predictions if it was larger and had additional features.

Reference

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